

## Artificial Intelligence Lab (AI 2002)

Date: March 17<sup>th</sup>, 2025

Course Instructor(s)  
Ms. Ramsha Jat

## Lab Mid Exam Paper B

Total Time: 90 minutes

Total Marks: 50

Total Questions: 03

Semester: Spring 2025

Campus: Karachi

Department: AI

### Instructions:

- The Exam paper consists of 3 questions on 2 pages.
- Submit ipynb files name under your nu id with question number. E.g: k22-xxxx\_Q1.ipynb.
- Submission will be on Software. Submission IP is **172.16.33.88:5000**.
- Enter your NU Id in Username and **Fast1234** in password.
- Read the question carefully before answering.

| Student Name | Roll No | Section | Student Signature |
|--------------|---------|---------|-------------------|
|--------------|---------|---------|-------------------|

### LLO # 1: [18 Marks]

**Q1.** A pirate agent is stranded on a grid-based island and must locate buried treasure. The island's terrain consists of water (W), land (L), rocks (R), and the treasure (T). The pirate can only move on land (L) and must navigate using breadth-First Search (BFS) to find the shortest path to the treasure. The agent's goal is to reach the treasure (T), and it cannot traverse water (W) or rock (R) cells. Identify the type of agent and implement BFS would be implemented to solve this problem. For the example input map below, show the path taken to reach the treasure.

```

L L L W T
L R L W W
L L L L W
R R L R W
L L L L L

```

### LLO # 3: [18 Marks]

**Q2.** Implement an A search algorithm\* to find the shortest path in an  $N \times M$  weighted grid, where each cell represents a terrain cost (a positive integer). Some cells are blocked (#), meaning they

Page 1 of

## National University of Computer and Emerging

cannot be traversed. The algorithm should determine the least-cost path from the top-left corner (0,0) to the bottom-right corner (N-1, M-1) while considering the terrain costs. The movement is restricted to four directions: up, down, left, and right, and the cost of movement is determined by the value of the destination cell. The heuristic function should use Manhattan distance to estimate the remaining cost to the goal. The implementation should efficiently handle larger grids and return both the optimal path and total travel cost.

```
1 1 2 # 3
1 # 2 # #
1 2 1 2 #
# # 1 # #
1 1 1 2 1
```

---

### LLO # 4: [14 Marks]

---

**Q3.** Implement a Genetic Algorithm (GA) to maximize the function

$$f(x) = x^2 - 1$$

within the range [0, 31].

Each solution should be represented as a 5-bit binary string, since  $2^5 = 32$ , covering all possible values of  $x$ . The algorithm should start by generating a random population of binary-encoded values. Each individual in the population should be evaluated based on its fitness, which is determined by computing  $f(x) = x^2 - 1$ . The algorithm should then apply natural selection, retaining only the fittest individuals based on their fitness scores. Next, selected parents should undergo single-point crossover, where a random point in their binary representation is chosen, and their bits are swapped to create new offspring. To introduce genetic diversity, each offspring should undergo single-point mutation, where a randomly chosen bit is flipped ( $0 \rightarrow 1$  or  $1 \rightarrow 0$ ) with a small probability. The process should continue for a fixed number of generations or until the population converges to an optimal solution. The algorithm should finally return the best  $x$  value found along with its corresponding  $f(x)$  value, demonstrating how GA effectively optimizes the function over successive generations